WEATHER CHAIN ANALYSIS ($D = 0$)

1. Introduction

In this report we will conduct a weather analysis for two regions, Region A and Region B. The analysis is based on two Markov models, Model 1 for Region A and Model 2 for Region B which both describe the transition between three weathers, namely Sunny (State 1), Overcast (State 2) and Rainy (State 3) between days. The weather is said to be in a Dry Spell if the weather Sunny occurs for two or more consecutive days, and likewise for Rainy Spell. The total number of days is called the duration of the Spell.

The aim is to provide recommendation to an agricultural firm on which region is more suitable on growing a new type of crop based on the following three optimal conditions:

- Condition 1: Probability of the weather Rainy occurs before Overcast, given that it is currently on a Dry Spell is at most 0.4.
- Condition 2: Long-term expected one-day weather proportion is close to $0.4, 0.25$ and 0.35 for Sunny, Overcast and Rainy day.
- Condition 3: Total expected rain intensity across a Rainy Spell is below 20.

Based on our analysis, we find that Region B is more suitable for growing the new crop as it attains all three conditions, where Region A attains none. We also provide some further consideration on the validity of the results.

2. Model Explanation

Under Model 1, the weather of any day is dependent on the previous day's weather. The probability of the next day's weather remains the same as current day as Sunny, Overcast, and Rainy are $p_{1,1} = \frac{7}{8}$, $p_{2,2} = \frac{15}{16}$ and $p_{3,3} = \frac{7}{8}$ respectively. Note that the probability of the weather remains as Overcast is comparatively higher. Otherwise, the weather will change to one of the two remaining weather with with equal probability.

Under Model 2, the weather of any day is dependent on the previous two days' weather. Regardless of the weather type, if the same weather occurs for two consecutive days, the probability of the next day's weather remains the same is $p = \frac{5}{12}$. Otherwise, the probability of transition to Sunny is always the highest, followed by Overcast, and then Rainy (See Appendix A.1). If the weather of two consecutive days are not the same, the weather will then change to Sunny, Overcast and Rainy with fixed probabilities of $\frac{1}{2}$, $\frac{1}{6}$ and $\frac{1}{3}$ respectively.

We noted that Model 2 is more superior than Model 1, as the former takes into account of an additional day's weather in predicting the next day's. However, this would lead to more complexity in our calculation.

For a Rainy day, its rain intensity is a measure of the duration and heaviness of the rain. For Region A, the rain intensity is exponentially distributed with rate $\frac{1}{3}$ whereas for Region B, it is uniformly distributed on the interval $(0, 10)$. Hence,

the expected intensity of a Rainy day for Region A and B is the mean of the distributions, which are 3 and 5 respectively.

The three optimal conditions can all be inferred from both models through simulations. Condition 1 can be studied by investigating the transition probabilities of the models, by comparing the likelihood of transition to Overcast and to Rainy once exiting from a Sunny Spell. Condition 2 can be studied by conducting a simulation with sufficiently large steps, and comparing the final weather proportion with the desirable proportion. Condition 3 can be studied by simulating Rainy Spells and obtaining the average duration. Relevant analytical calculations will be provided in Appendix to support our justification.

3. Results

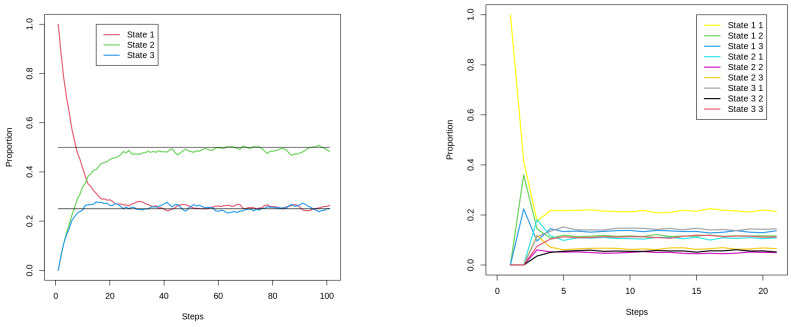
The probability in Condition 1 can be interpreted as the probability of transition to Rainy (rather than Overcast) once exiting a Dry Spell. The condition of being in a Dry Spell, can be interpreted as being in a Sunny day for Model A, and as being in two consecutive Sunny days for Model B due to the Markov property. We summarise the simulation and theoretical result in Table 1. More details of the latter can be found in Appendix A.2.

Region	A	B
Simulation result	0.498	0.409
Theoretical result	0.5	0.4

Table 1. A comparison of Region A and B for Condition 1

We noted that for Region A, the probability exceeds the 0.40 threshold. For Region B, whilst our simulation result was slightly beyond 0.40, the theoretical result is founded to be exactly the threshold. Hence, we conclude that Region B satisfies Condition 1, whereas Region A does not.

Next, Condition 2 is studied through running a simulation of 100 days and 20 days for Model 1 and 2 respectively, starting with a Sunny day for Region A, and two consecutive Sunny days for Region B. The results are depicted in Figure 1.



(a) Model 1 (b) Model 2

Figure 1. Long-term proportion of one-day weather using Model 1, and two-days weather using Model 2.

Note that the rate of convergence of Model 1 is slower than of Model 2. The long-term two-days weather proportion from Model 2 can be further post-processed into of one-day weather, shown in Figure 2.

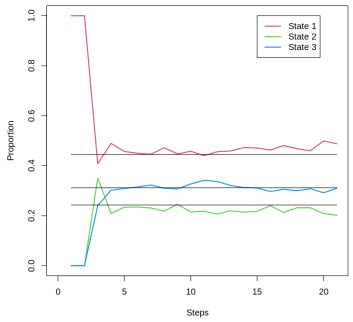


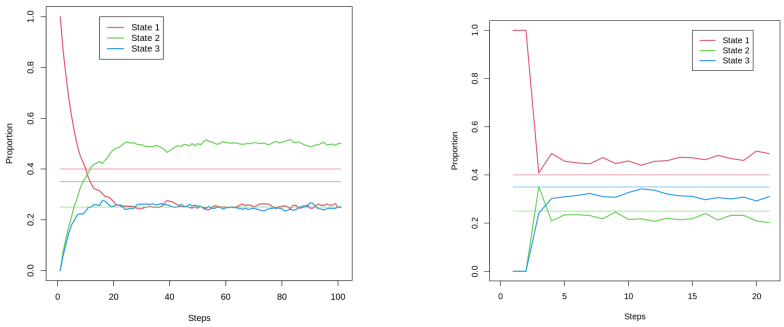
FIGURE 2. Long-term one-day weather proportion of Region B

The long-term one-day weather proportion of both models are independent on the initial weather(s), and they must coincide with the stationary distribution (represented as black horizontal lines in Figure 1(A) and Figure 2) of their respective models (See Appendix A.3). The simulated results are given in Table 2.

Region	A	B
Long-term proportion of weather	$(0.251, 0.515, 0.235)$	$(0.451, 0.222, 0.327)$

TABLE 2. A comparison of Region A and B for Condition 2

A comparison with the desirable proportion in Condition 2 is shown in Figure 3.



(A) Region A (B) Region B

FIGURE 3. A comparison between long-term weather proportion of each region and the desirable proportion $(0.4, 0.25, 0.35)$.

The result clearly shows that Region B has better proximity to the optimal weather proportion than Region A. We conclude that for Condition 2, Region B performs better than Region A.

Lastly, for Condition 3, the total expected intensity across a Rainy spell can be obtained by multiplying the expected duration of a Rainy spell with the expected rain intensity of a Rainy day. The results are summarised in Table 3.

Region	A	B
Expected duration of a Rainy Spell (days)	8.93	2.73
Expected rain intensity of a Rainy day	3	5
Total expected intensity across a Rainy Spell	26.7	13.7

TABLE 3. A comparison of Region A and B for Condition 3

Region A's total expected intensity across a Rainy Spell lies beyond the threshold of 20 mainly due to its long expected duration of Rainy Spell. Despite the expected rain intensity of a Rainy day in Region B is higher, the much shorter expected duration of its Rainy Spell resulted in its total expected intensity across a Rainy Spell to fall within the desirable range. See exact calculation in Appendix A.4. We conclude that Region B satisfies Condition 3, whereas Region A does not.

4. Conclusion

The results indicated that all three optimal conditions were attained by Region B. In contrast, Region A failed to perform better in any of the conditions. We would therefore recommend Region B for growing the new crop.

As a point of consideration, constructing a two-day weather model for Region A would lead to a more valid comparison between the regions. However, it is anticipated that the high probability of its weather remaining overcast would render it unsuitable for growing the crop anyway. Finally, the validity period of the models' results requires further research due to the seasonal nature of weather.

Appendix A. Outline of calculations

A.1. **Markov chain models.** Transition matrix of Model 1 and Model 2 are P_1 and P_2 respectively, as follows:

$$P_1 = \begin{pmatrix} \frac{7}{8} & \frac{1}{16} & \frac{1}{16} \\ \frac{1}{32} & \frac{15}{16} & \frac{1}{32} \\ \frac{1}{16} & \frac{1}{16} & \frac{7}{8} \end{pmatrix} \text{ and } P_2 = \begin{pmatrix} \frac{5}{12} & \frac{7}{20} & \frac{7}{30} & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & \frac{1}{2} & \frac{1}{6} & \frac{1}{3} & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & \frac{1}{2} & \frac{1}{6} & \frac{1}{3} \\ \frac{1}{2} & \frac{1}{6} & \frac{1}{3} & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & \frac{7}{16} & \frac{5}{12} & \frac{7}{48} & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & \frac{1}{2} & \frac{1}{6} & \frac{1}{3} \\ \frac{1}{2} & \frac{1}{6} & \frac{1}{3} & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & \frac{1}{2} & \frac{1}{6} & \frac{1}{3} & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & \frac{7}{36} & \frac{7}{18} & \frac{5}{12} \end{pmatrix}$$

- Calculation of transition probabilities for Model 1:
  - $p_{1,1} = 1 - \frac{50}{400(2-(1-2)^2)} = \frac{7}{8}$. Similarly, $p_{2,2} = \frac{15}{16}$ and $p_{3,3} = \frac{7}{8}$.
  - For $i \neq j$, $p_{i,j} = \frac{1-p_{i,i}}{2}$.
- Calculation of transition probabilities for Model 2:
  - $p_{(i,i),(i,i)} = p = \frac{5}{12}$ for $i = 1, 2, 3$.

- $p_{(1,1),(1,2)} = \frac{7}{12}\left(1 - \frac{2}{2+3}\right) = \frac{3}{5}(1-p)$ and $p_{(1,1),(1,3)} = \frac{2}{5}(1-p)$. Similarly, $p_{(2,2),(2,1)} = \frac{3}{4}(1-p)$, $p_{(2,2),(2,3)} = \frac{1}{4}(1-p)$, $p_{(3,3),(3,1)} = \frac{2}{3}(1-p)$ and $p_{(3,3),(3,2)} = \frac{1}{3}(1-p)$.
- Let $i \neq j$. For $k = 1, 2, 3$, $p_{(i,j),(j,k)}$ are $\frac{1}{2}$, $\frac{1}{6}$ and $\frac{1}{3}$ respectively.
- For $j \neq k$, $p_{(i,j),(k,l)} = 0$.

Note that the underlying one-day weather model of Model 2 does not satisfy the Markov property.

A.2. **Hitting probabilities.** Due to the Markov property, the probability of the weather Rainy occurs before Overcast, given that it is currently on a Dry Spell can be interpreted as follows:

- Model 1: Hitting probability of State 3 before State 2, given starting at State 1. As the Markov chain must hit one of State 2 and State 3 once leaving State 1, the probability is $\frac{p_{13}}{p_{13}+p_{12}} = 0.5$.
- Model 2: Hitting probability of $(1,3)$ before $(1,2)$, given starting at $(1,1)$. As the Markov chain must hit one of $(1,2)$ and $(1,3)$ once leaving $(1,1)$, the probability is $\frac{p_{(1,1),(1,3)}}{p_{(1,1),(1,2)}+p_{(1,1),(1,3)}} = 0.4$.

A.3. **Long term proportion of weathers.** For both models, the Markov chains are of finite state space and irreducible. In addition, they are aperiodic as both P_1 and P_2 has atleast one positive diagonal entries, e.g., p_{11} and $p_{(1,1),(1,1)}$. Hence, the limiting distribution associated to any initial distribution exists and must coincide with the unique stationary distribution. Let the stationary distribution of Model 1 and 2 be $\boldsymbol{\pi}_A = (\pi_i)_{i\in S_1}$ and $\boldsymbol{\pi}_B = (\pi_{(i,j)})_{(i,j)\in S_2}$. Solve the stationarity equations $\boldsymbol{\pi}_A P_1 = \boldsymbol{\pi}_A$ and $\sum_{i\in S_1} \pi_i = 1$ give:

$$\boldsymbol{\pi}_A = (0.25, 0.50, 0.25).$$

Solving the stationarity equations $\boldsymbol{\pi}_B P_2 = \boldsymbol{\pi}_B$ and $\sum_{(i,j)\in S_2} \pi_{(i,j)} = 1$ give:

$$\boldsymbol{\pi}_B = (0.2053, 0.1118, 0.1277, 0.1181, 0.0540, 0.0709, 0.1214, 0.0772, 0.1135).$$

Further, for Model 2, note that the long term proportion of one-day weather in State i is $\sum_{j\in S_1} \pi_{(i,j)}$ for $i = 1, 2, 3$, giving the distribution $(0.445, 0.243, 0.312)$.

A.4. **Expected duration of a rainy spell.** Define the random variable $T_{RS,A}$ and $T_{RS,B}$ to be the duration of a Rainy Spell in Region A and B respectively.

We can write $T_{RS,A} = 2 + T_3$, where T_3 is the expected number of days to stay at Rainy, given its currently Rainy in Region A. Note that:

$$\mathbf{E}T_3 = \sum_{k\geq 0} k p_{33}^k (1-p_{33}) = \sum_{k\geq 1} k p_{33}^k (1-p_{33}) = p_{33} \sum_{k\geq 1} k p_{33}^{k-1}(1-p_{33}) = \frac{p_{33}}{1-p_{33}}.$$

where the final equality follows from that the summation is precisely the expectation of a geometric random variable with parameter $1 - p_{33}$. This gives:

$$\mathbf{E}T_{RS,A} = 2 + \mathbf{E}T_3 = 2 + \frac{p_{33}}{1-p_{33}} = 2 + \frac{7/8}{1-7/8} = 9 \,\text{days}.$$

By a similar argument, write $T_{RS,B} = 2 + T_{(3,3)}$, where $T_{(3,3)}$ is the expected number of days to stay at $(3, 3)$, given its current two-days state is $(3, 3)$ in Region B. Hence:

$$\mathbf{E}T_{RS,B} = 2 + \mathbf{E}T_{(3,3)} = 2 + \frac{p_{(3,3)}}{1 - p_{(3,3)}} = 2 + \frac{5/12}{1 - 5/12} \approx 2.71 \text{ days.}$$